DG Interns Hub – Cybersecurity Internship

Week 7 Assignment Report: VAPT Web Application with Burp Suite

Author: B Sai Amulya Goud

Intern Id: DG/SEPTEMBER/CYBER/179

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OBJECTIVES

The main objective of this assignment is to understand and perform basic web application security testing using Burp Suite Community Edition in a controlled laboratory environment.

The specific objectives are:

1. **Burp Suite Setup and Configuration:** To install Burp Suite Community Edition and configure the browser proxy for intercepting HTTP and HTTPS traffic.
2. **HTTP Request Interception:** To capture, inspect, and analyze HTTP requests and responses, including login requests and session-related information.
3. **Web Application Reconnaissance:** To explore the target application and identify accessible pages, directories, endpoints, login forms, and input fields.
4. **Application Mapping:** To use Burp Suite's Target Site map and HTTP history features to understand the structure and attack surface of the web application.
5. **Vulnerability Identification:** To investigate common web application vulnerabilities, including SQL Injection, Cross-Site Scripting (XSS), authentication weaknesses, and parameter tampering.
6. **Manual Security Testing:** To use Burp Suite tools such as Proxy and Repeater to modify requests, compare server responses, and verify suspected vulnerabilities.
7. **Directory and Endpoint Discovery:** To identify accessible and potentially hidden application resources using Burp Suite
8. **Evidence Collection:** To capture screenshots of relevant requests, responses, browser behavior, and testing results to support vulnerability findings.
9. **Vulnerability Assessment and Reporting:** To document identified vulnerabilities with descriptions, reproduction steps, evidence, impact, severity, and remediation recommendations.
10. **Security Awareness and Best Practices:** To understand common web application security risks and recommend appropriate mitigation techniques to improve application security.
11. **Practical Skill Development:** To gain hands-on experience in web application penetration testing, HTTP traffic analysis, attack-surface mapping, and security report preparation.

All testing activities are performed against intentionally vulnerable applications, such as DVWA, within an authorized and controlled laboratory environment.

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TASK 22 BURP SUITE SETUP & INTERCEPTION

The objective of this task is to set up Burp Suite Community Edition for web application security testing and understand how HTTP and HTTPS requests and responses can be intercepted and analyzed.

Introduction to Burp Suite

Burp Suite is a web application security testing platform used by penetration testers and security professionals to identify vulnerabilities in web applications.

Browser Proxy Configuration

A proxy server acts as an intermediary between a web browser and a web server. For this task, the browser is configured to use the local proxy address `127.0.0.1` and port `8081`. This allows Burp Suite to receive and inspect browser traffic.

```
curl -I --proxy http://127.0.0.1:8081 http://neverssl.com
```

HTTP and HTTPS Interception

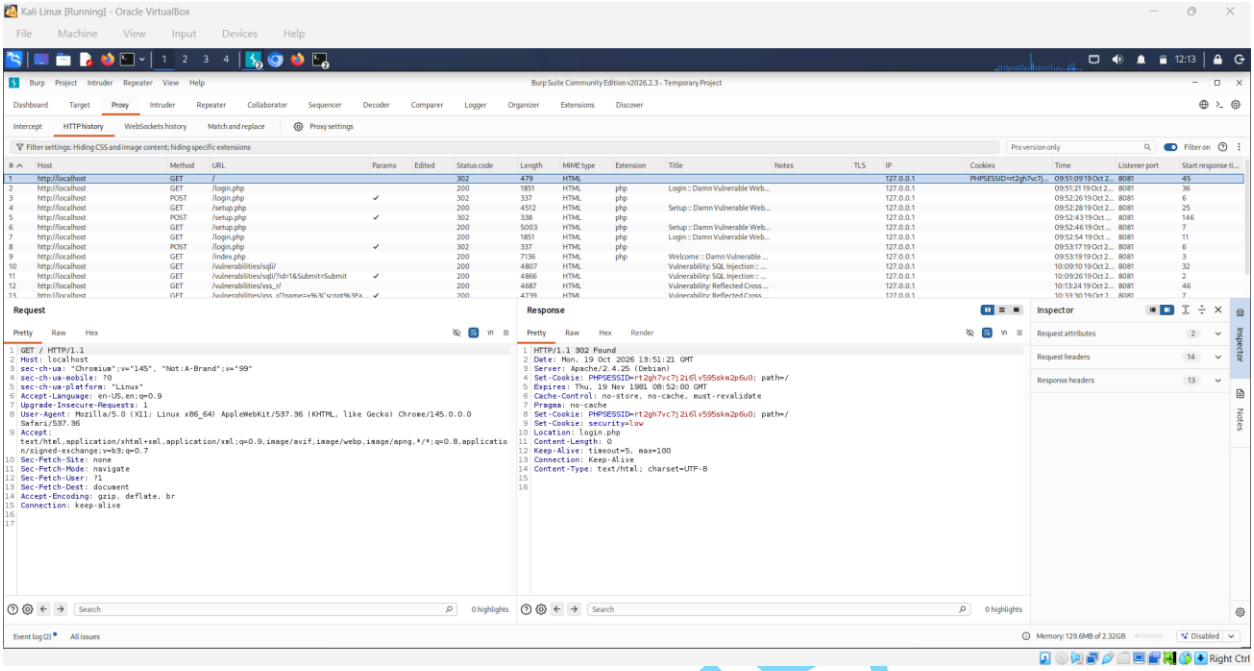
HTTP is a protocol used for communication between web browsers and servers.

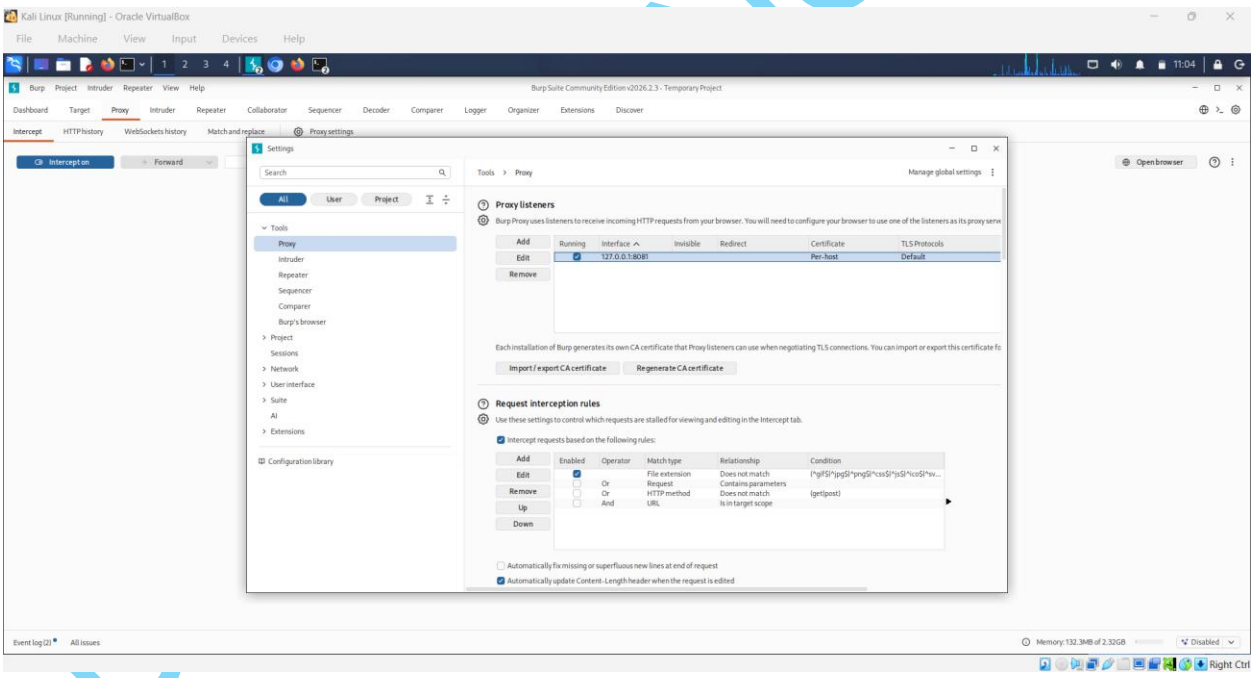
HTTPS protects communication using TLS encryption. To inspect HTTPS traffic, Burp Suite uses its own Certificate Authority (CA) certificate, which must be trusted by the testing browser.

Tools and Technologies

- Burp Suite Community Edition
- Kali Linux
- Mozilla Firefox
- HTTP and HTTPS protocols
- Burp Suite Proxy and CA certificate

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Expected Outcome

After completing this task, the tester should be able to configure a browser to communicate through Burp Suite, intercept and inspect HTTP requests, understand the requirements for HTTPS interception, and analyze a login request in an authorized practice application.

TASK 23 Reconnaissance and Web Application Mapping

Burp Suite Community Edition was launched in Kali Linux. The browser was configured to route web traffic through the Burp proxy at `127.0.0.1:8081`.

The target application, DVWA, was accessed locally through the browser. This ensured that reconnaissance was performed within the controlled training environment.

Capture HTTP Requests:

The Proxy and HTTP history features of Burp Suite were used to capture requests generated while browsing the application.

Each request was examined to identify the requested URL, HTTP method, parameters, response status code, and other relevant details.

Analyze HTTP Methods and Parameters

The captured requests were analyzed to distinguish between GET and POST methods.

- **GET:** Commonly used to retrieve resources or submit parameters through the URL.
- **POST:** Commonly used to submit form data in the request body.

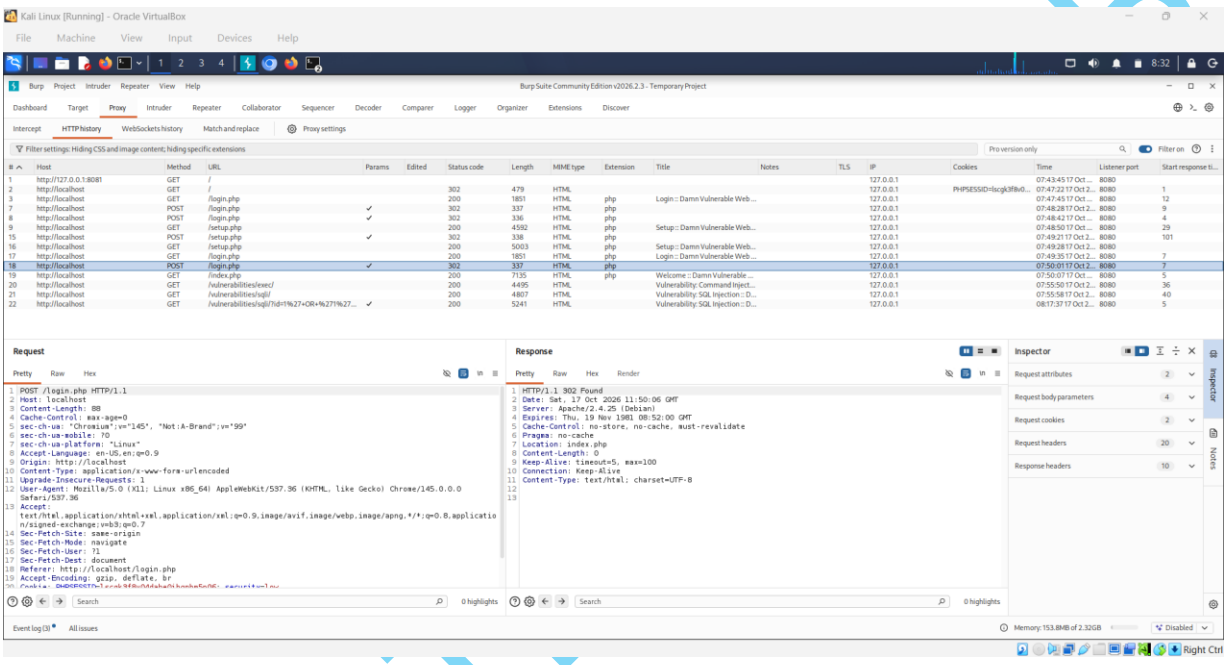
Document the Attack Surface

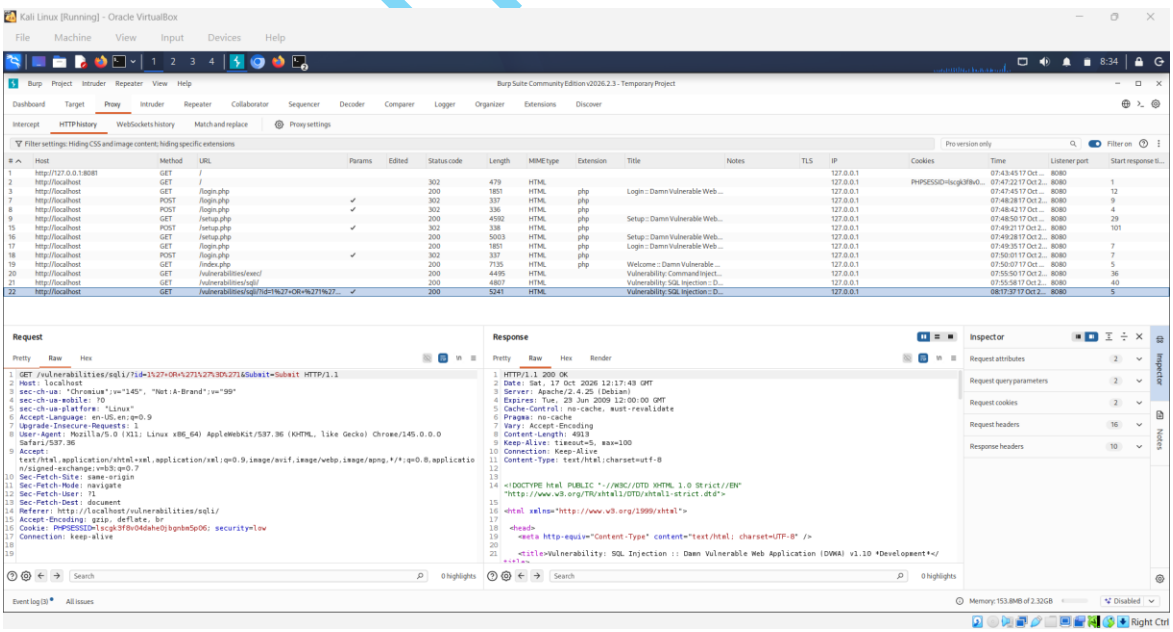
The discovered endpoints, HTTP methods, parameters, and application functions were organized into an attack-surface inventory.

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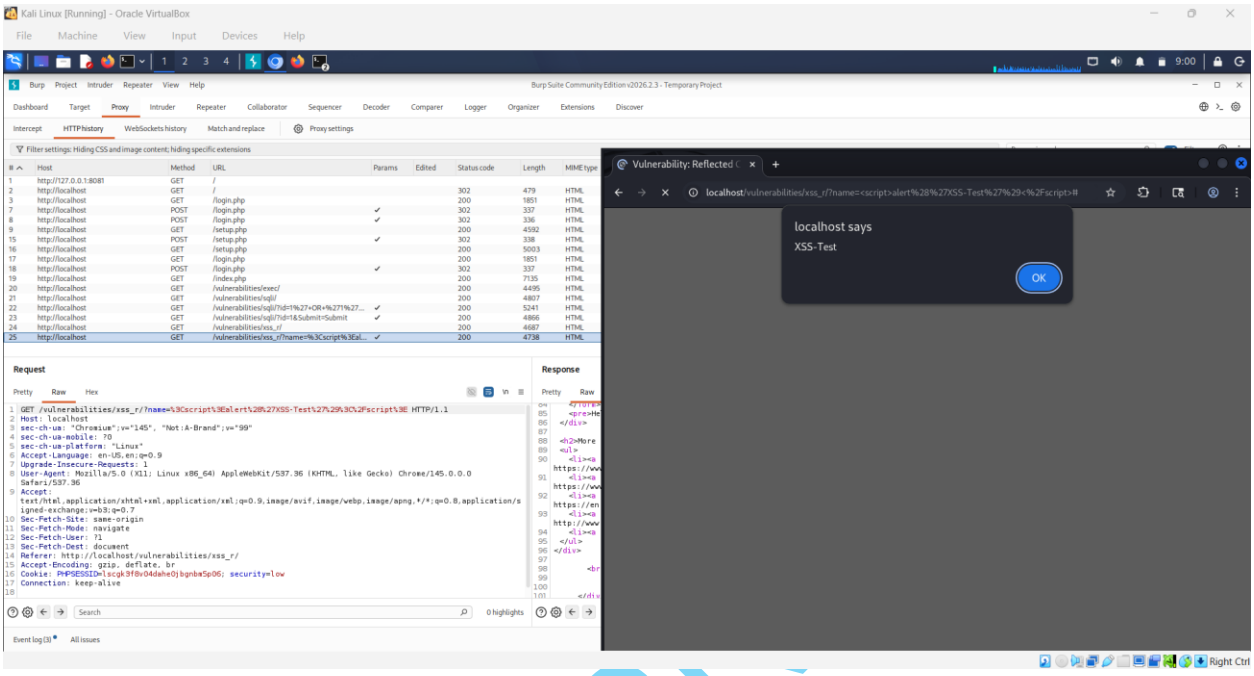
This inventory provides a foundation for subsequent vulnerability testing, including SQL injection, cross-site scripting, authentication testing, and parameter tampering.

SQL Injection:

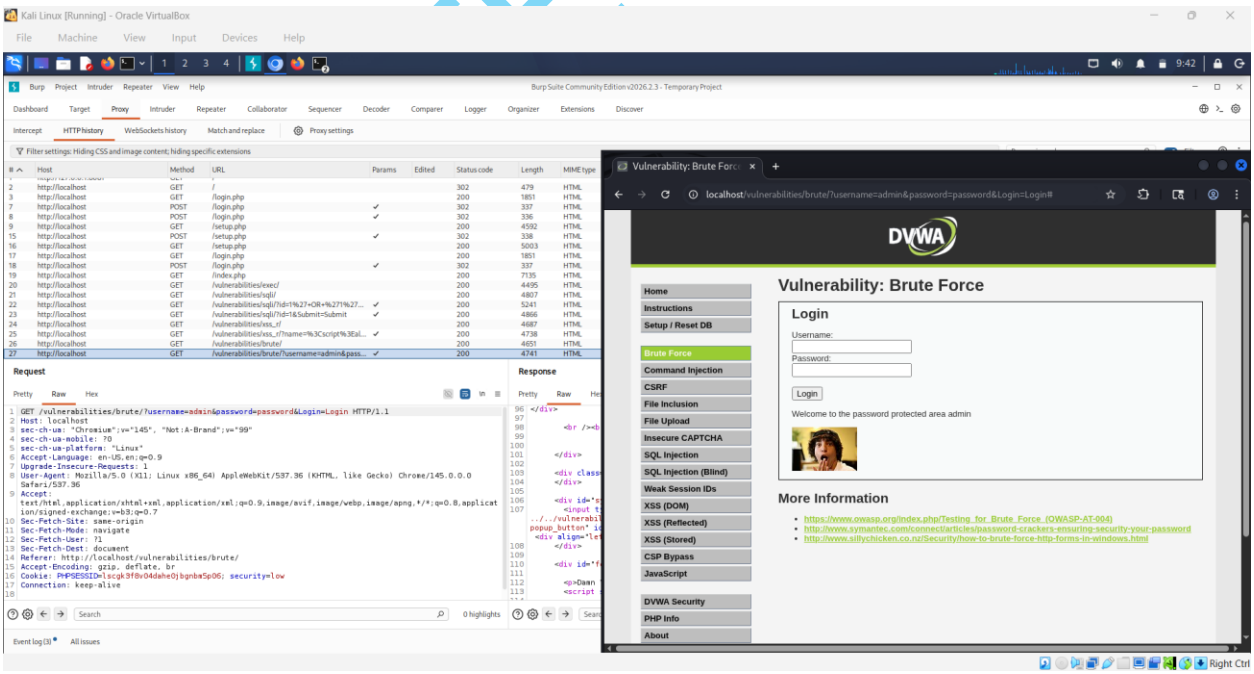


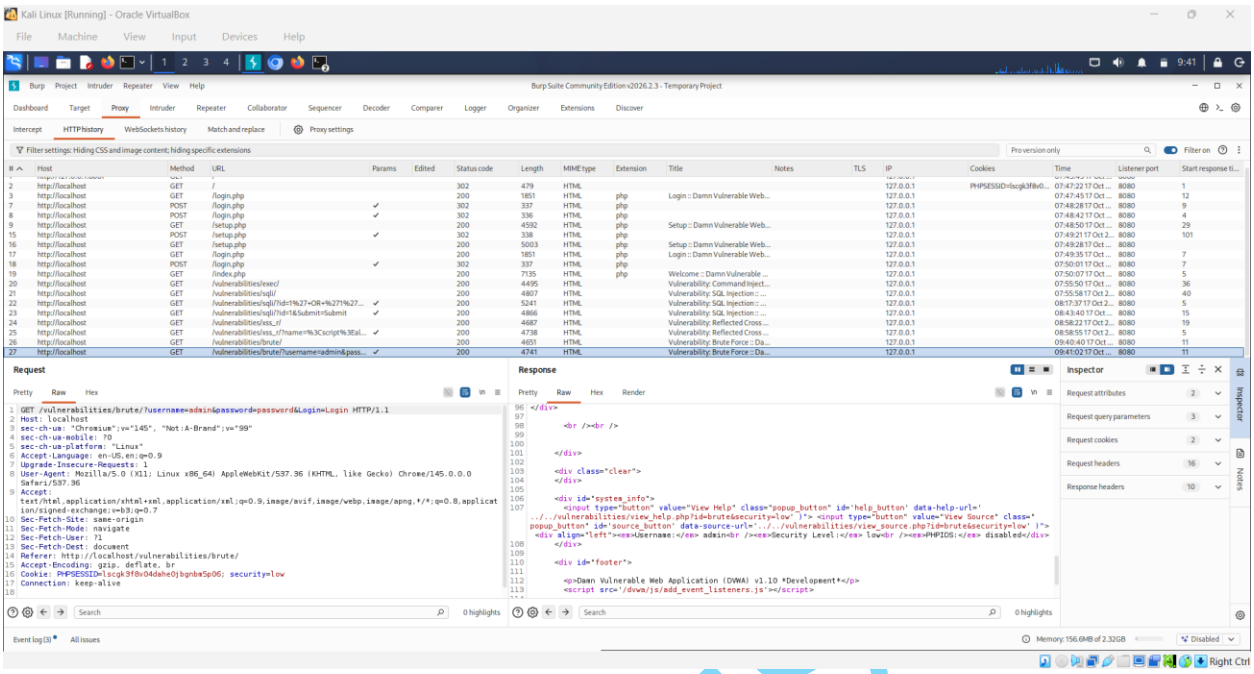


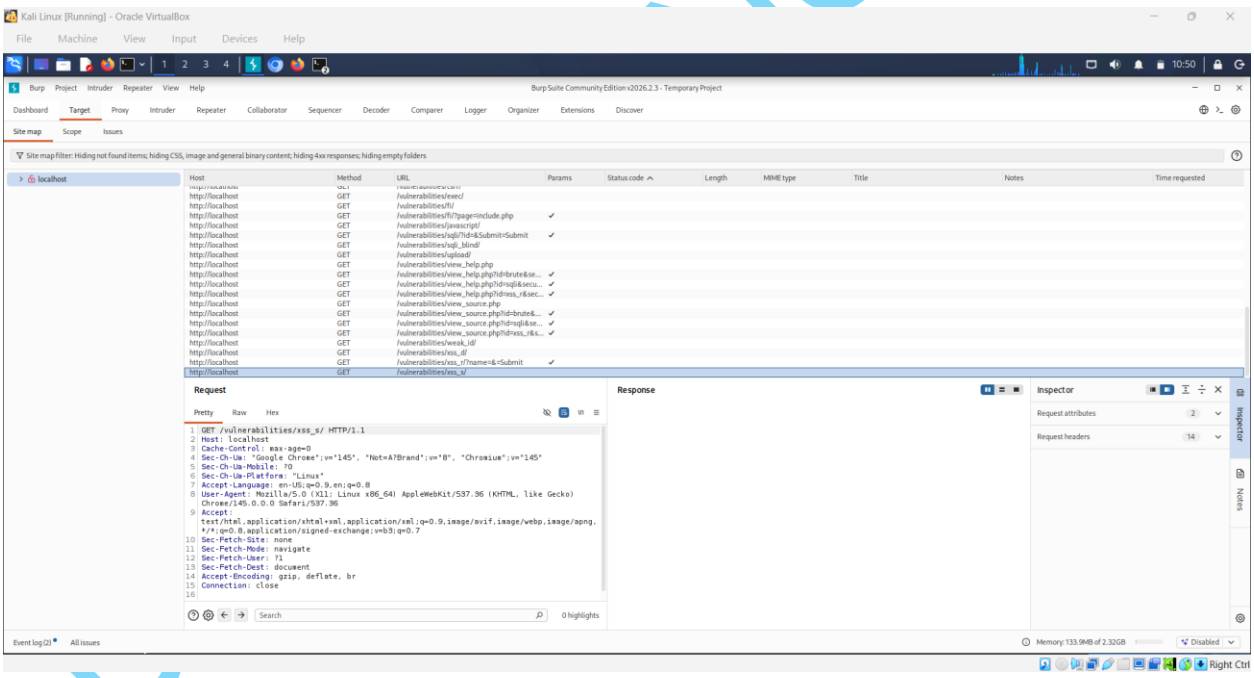
XSS (Reflected):



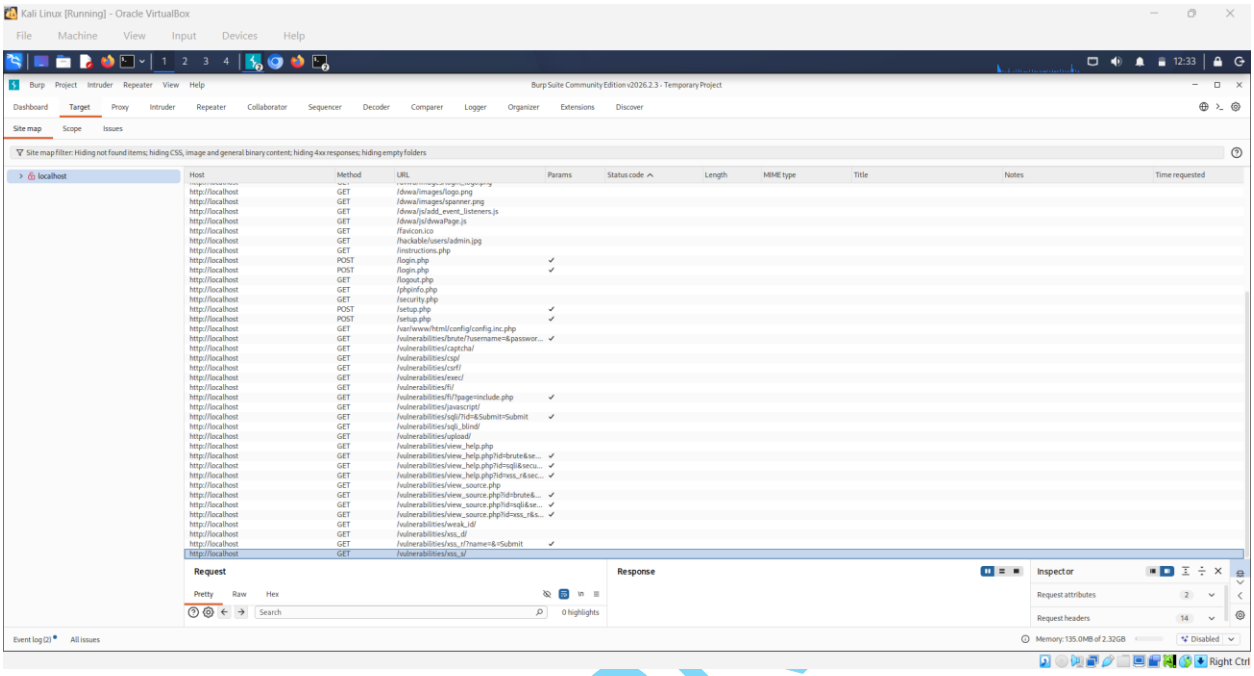
Brute:




Kali Linux [Running] - Oracle VirtualBox
Burp Suite Community Edition v2026.2.3 - Temporary Project
Request
Response
Inspector


Kali Linux [Running] - Oracle VirtualBox
Burp Suite Community Edition v2026.2.3 - Temporary Project
Request
Response
Inspector

Sitemap:



Observed Endpoint	Method shown	Purpose
`/instructions.php`	GET	Instructions page
`/login.php`	POST	Login form submission
`/logout.php`	GET	Logout endpoint
`/setup.php`	GET and POST	Setup page and form submission
`/vulnerabilities/brute/`	GET	Brute Force module

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`/vulnerabilities/csrf/`	GET	CSRF module
`/vulnerabilities/exec/`	GET	Command Injection module
`/vulnerabilities/fi/`	GET	File Inclusion module
`/vulnerabilities/sqli/`	GET	SQL Injection module
`/vulnerabilities/upload/`	GET	File Upload module
`/vulnerabilities/xss_r/`	GET	Reflected XSS module
`/vulnerabilities/xss_s/`	GET	Stored XSS module

Observation and Results:

The Burp Suite Site map and HTTP history showed multiple accessible DVWA endpoints. These included authentication pages, application navigation pages, setup pages, and intentionally vulnerable training modules.

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TASK 24 Vulnerability Testing

AIM:

To identify and analyze common web application vulnerabilities using Burp Suite Community Edition and the Damn Vulnerable Web Application (DVWA) in a controlled laboratory environment.

Tools and Technologies

- Burp Suite Community Edition: Intercepts and analyzes HTTP/HTTPS traffic.
- DVWA: An intentionally vulnerable web application used for security practice.
- Kali Linux: The testing environment.
- Web browser: Accesses the local DVWA application.

Types of Vulnerability Testing:

SQL Injection:

Input: 1' OR '1'='1
Output: ID: 1' OR '1'='1

```
First name: admin

Surname: admin

ID: 1' OR '1'='1

First name: Gordon

Surname: Brown

ID: 1' OR '1'='1
```

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First name: Hack

Surname: Me

ID: 1' OR '1'='1

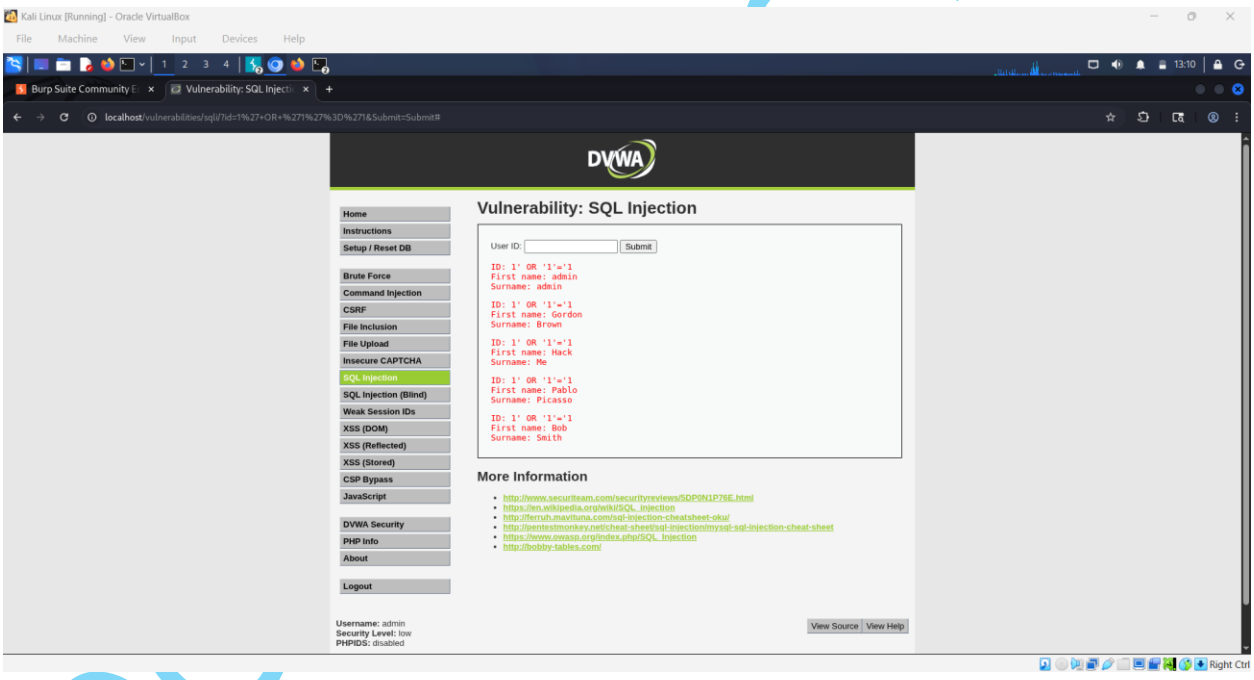
First name: Pablo

Surname: Picasso

ID: 1' OR '1'='1

First name: Bob

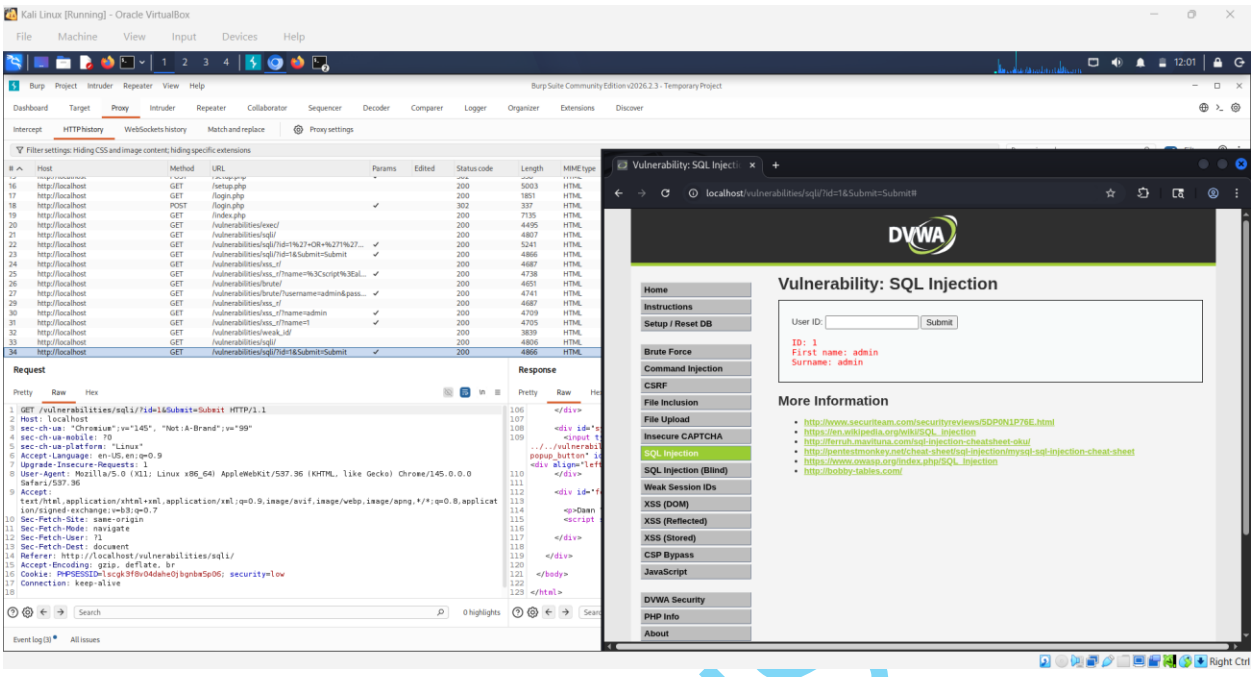
Surname: Smith



Input: 1

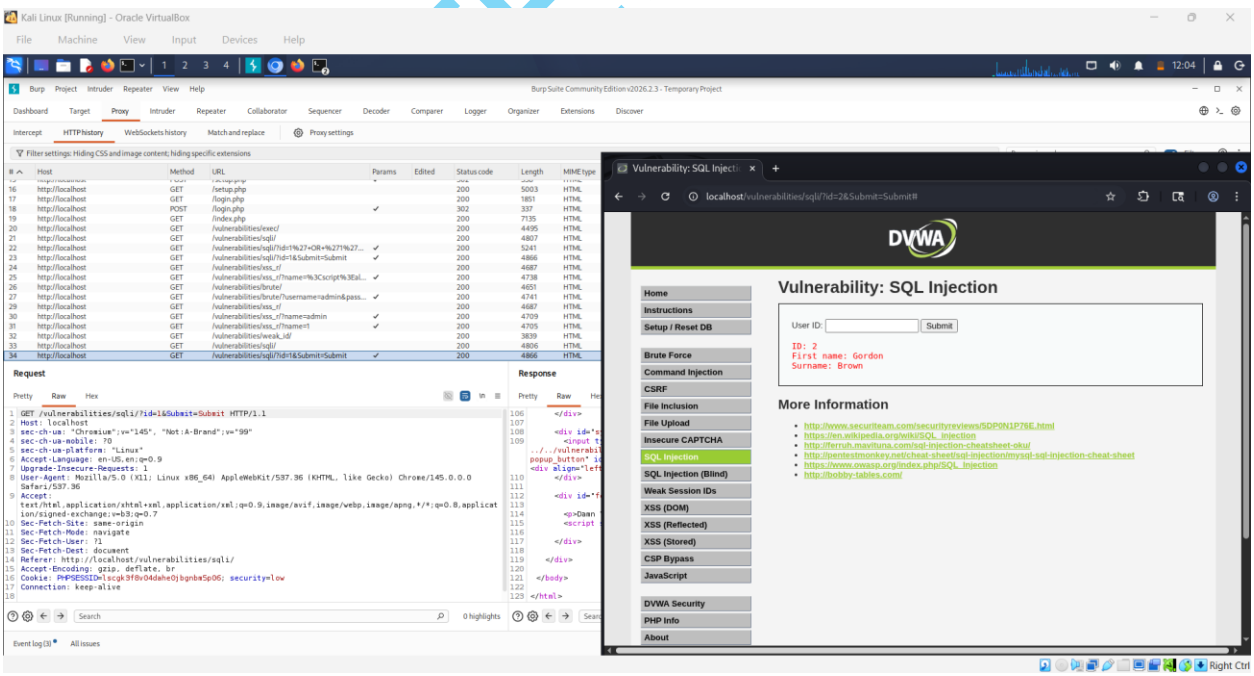
Output: first name: admin

Surname: admin



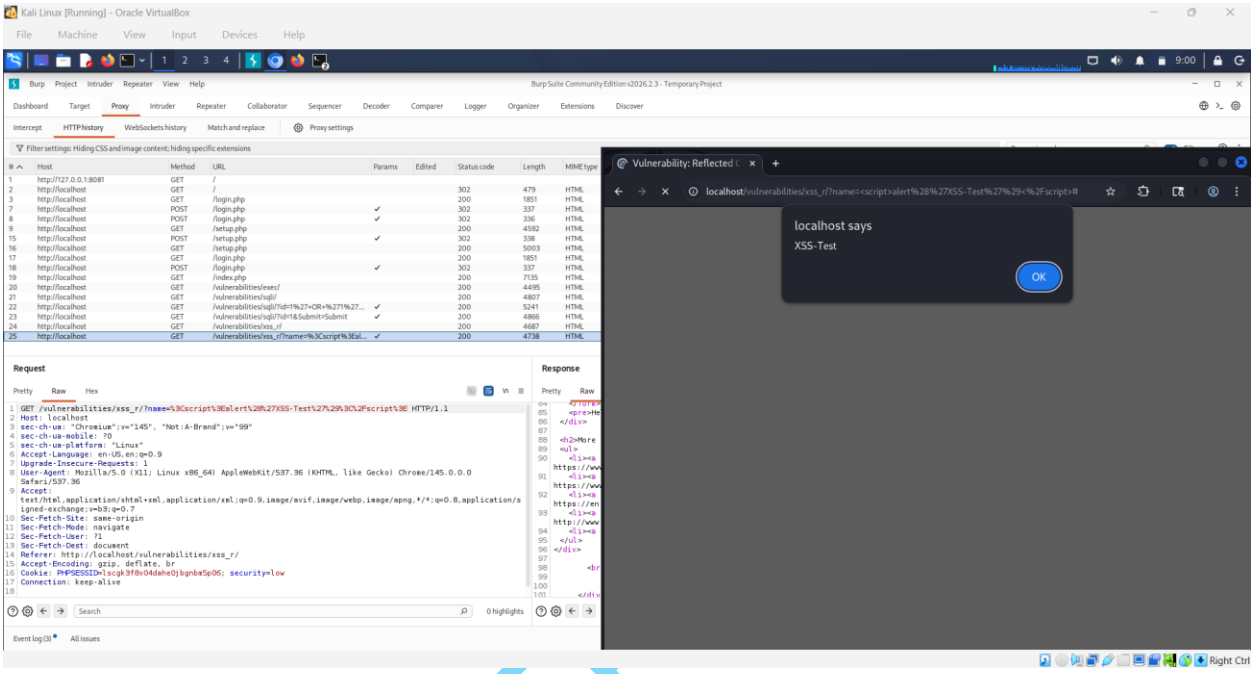
Input: 2

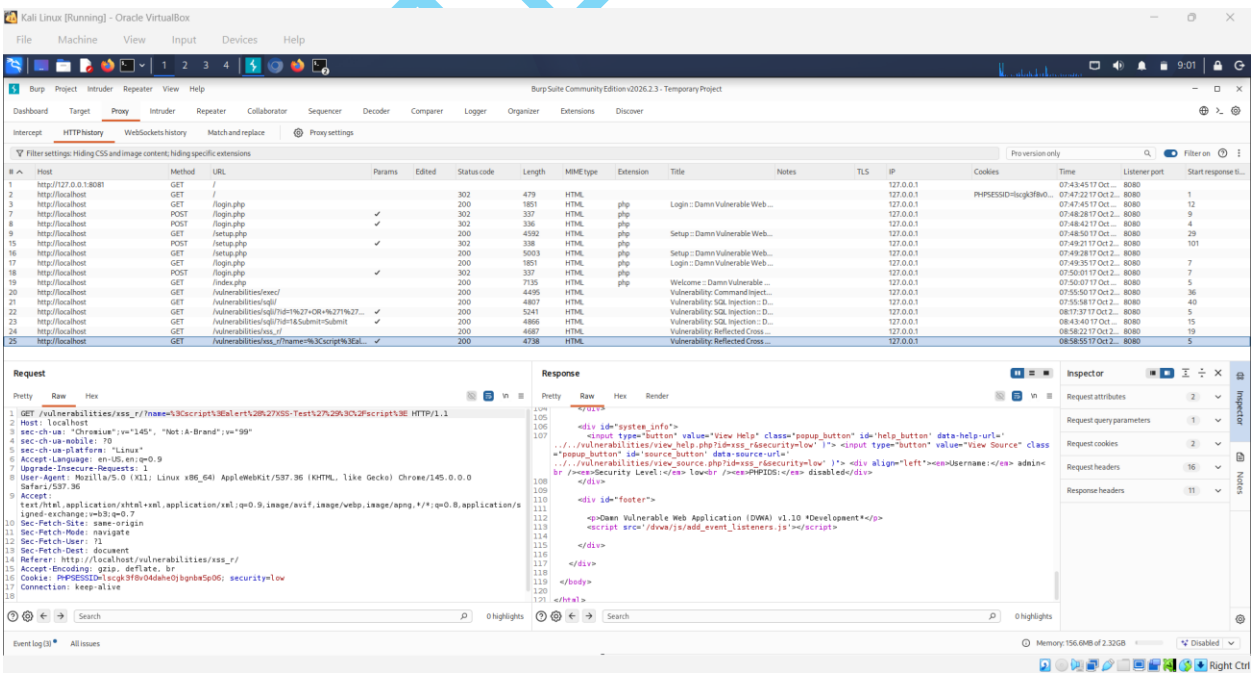
Output: first name: Gordon
Surname: Brown



XSS (Reflected):

Input: <script>alert('XSS-Test')</script>
Output : XSS-Test

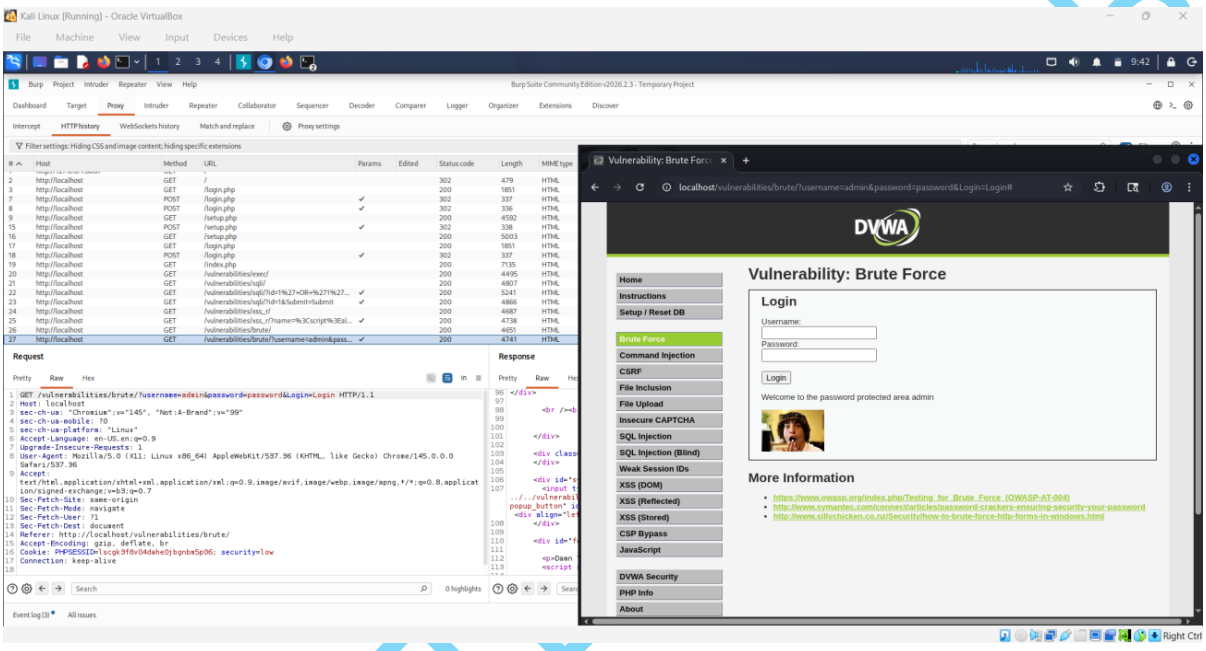




Brute:

Input: Username: admin
Password: password

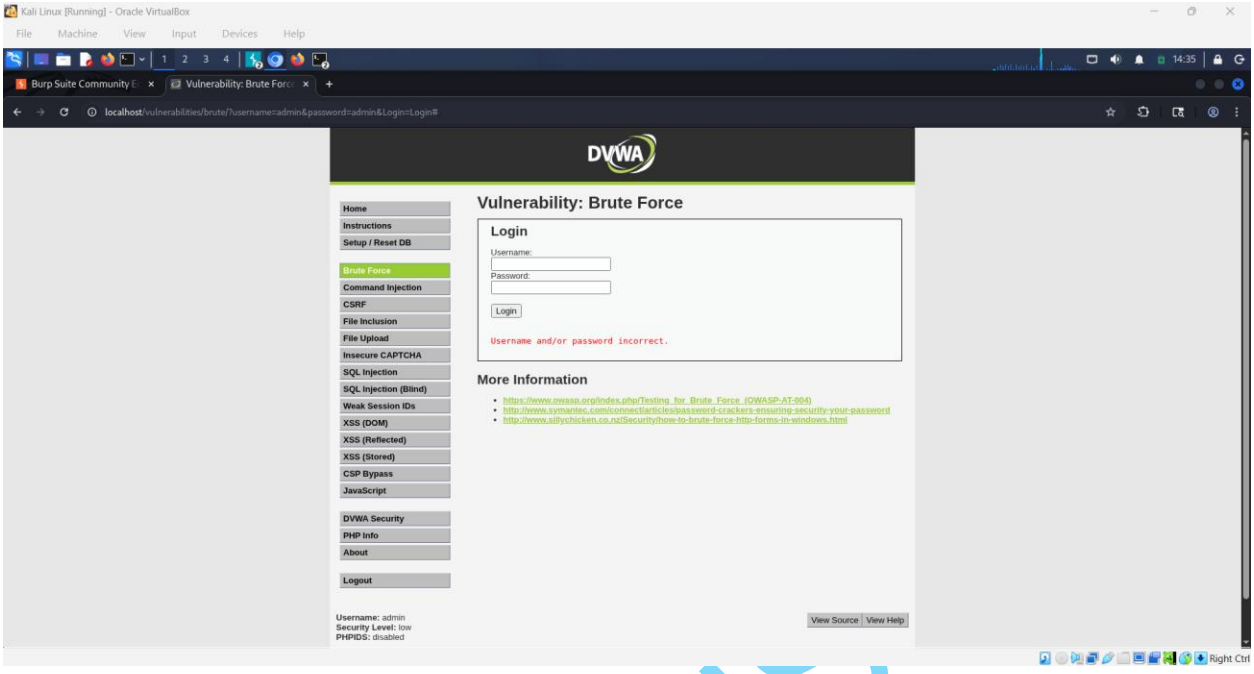
Output: Welcome to the password protected area admin

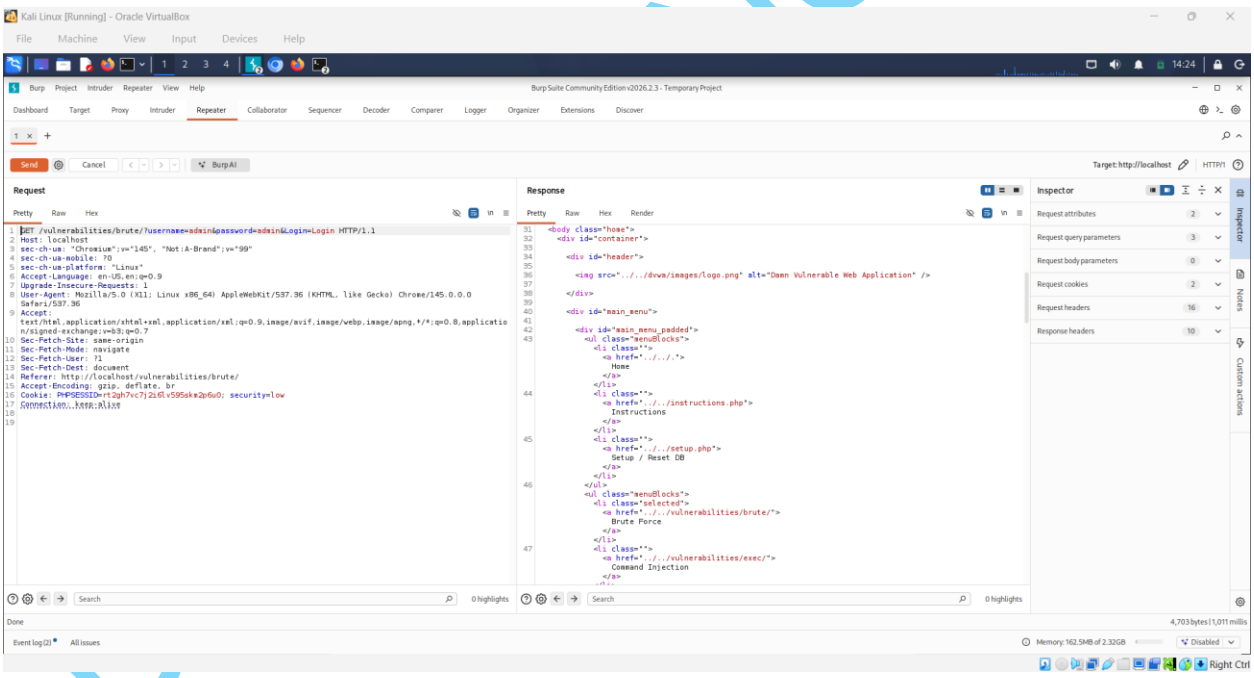


INPUT: Username: admin
Password: admin

OUTPUT: Username and/or password incorrect.

DG INT


Kali Linux [Running] - Oracle VirtualBox
File Machine View Input Devices Help
Burp Suite Community E
Vulnerability: Brute Forc
localhost/vulnerabilities/brute/?username=admin&password=admin&Login=Login#
DVWA
Home
Instructions
Setup / Reset DB
Brute Force
Command Injection
CSRF
File Inclusion
File Upload
Insecure CAPTCHA
SQL Injection
SQL Injection (Blind)
Weak Session IDs
XSS (DOM)
XSS (Reflected)
XSS (Stored)
CSP Bypass
JavaScript
DVWA Security
PHP Info
About
Logout
Vulnerability: Brute Force
Login
Username:
Password:
Login
Username and/or password incorrect.
More Information
https://www.owasp.org/index.php/Testing_for_Brute_Force_(OWASP-AT-004)
http://www.symantec.com/connect/articles/password-crackers-ensuring-security-your-password
http://www.sillychicken.co.nz/Security/how-to-brute-force-http-forms-in-windows.html
Username: admin
Security Level: low
PHPIDS: disabled
View Source
View Help
Right Ctrl


Kali Linux [Running] - Oracle VirtualBox
File Machine View Input Devices Help
Burp Project Intruder Repeater View Help
Burp Suite Community Edition v2026.2.3 - Temporary Project
Dashboard Target Proxy Intruder Repeater Collaborator Sequencer Decoder Comparer Logger Organizer Extensions Discover
Send
Cancel
Target: http://localhost
HTTP/1
Request
Pretty Raw Hex
GET /vulnerabilities/brute/?username=admin&password=admin&Login=Login HTTP/1.1
Host: localhost
sec-ch-ua: "Chromium";v="145", "Not:A-Brand";v="99"
sec-ch-ua-mobile: ?0
sec-ch-ua-platform: "Linux"
Accept-Language: en-US,en;q=0.9
Upgrade-Insecure-Requests: 1
User-Agent: Mozilla/5.0 (X11; Linux x86_64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/145.0.0.0 Safari/537.36
Accept: text/html,application/xhtml+xml,application/xml;q=0.9,image/avif,image/webp,image/apng,*/*;q=0.8,application/signed-exchange;v=b3;q=0.7
Sec-Fetch-Site: same-origin
Sec-Fetch-Mode: navigate
Sec-Fetch-User: ?1
Sec-Fetch-Dest: document
Referer: http://localhost/vulnerabilities/brute/
Accept-Encoding: gzip, deflate, br
Cookie: PHPSESSID=rt2gh7vc7j2i6lv595skm2p6u0; security=low
Connection: keep-alive
Response
Pretty Raw Hex Render
<body class="hose">

<ul class="menuBlocks">
<li class="">

Home

<li class="">

Instructions

<li class="">

Setup / Reset DB

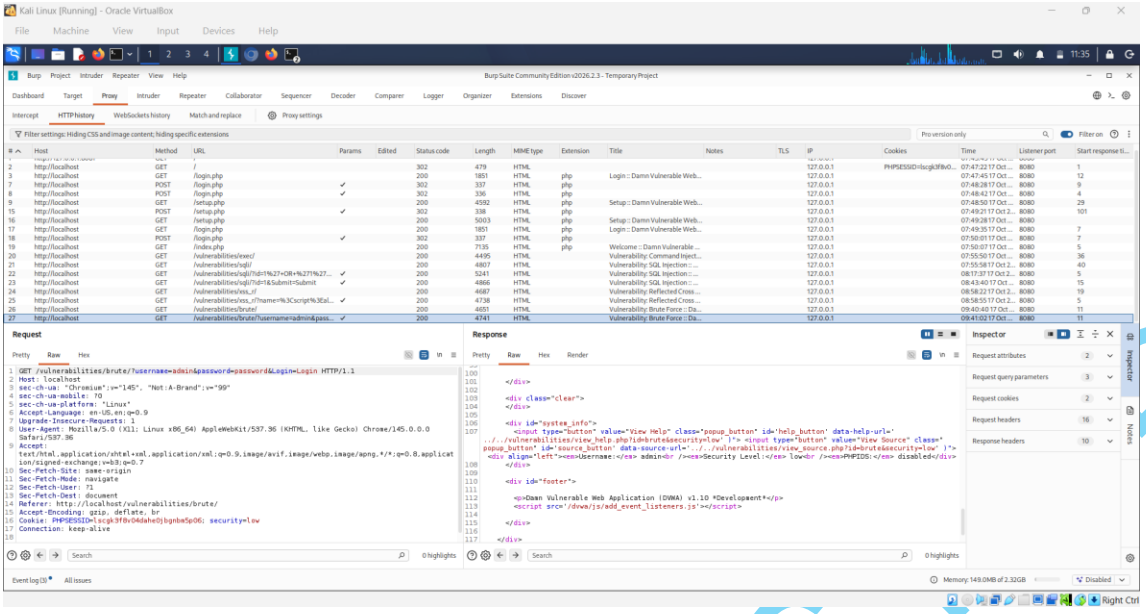
<ul class="menuBlocks">
<li class="selected">

Brute Force

<li class="">

Command Injection

Inspector
Request attributes
Request query parameters
Request body parameters
Request cookies
Request headers
Response headers
Search
0 highlights
Done
4,703 bytes | 1,011 millis
Event log (2)
All issues
Memory: 162.5MB of 2.32GB
Disabled
Right Ctrl



Result and Observation:

During the assessment of the local DVWA application, Burp Suite was used to capture requests and map accessible application paths.

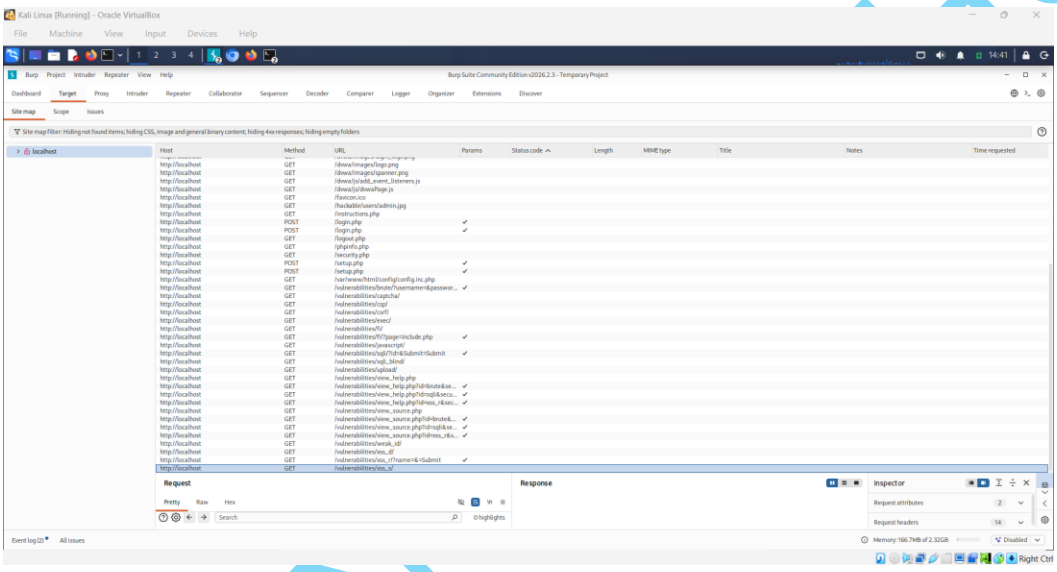
The Site map showed authentication pages, application pages, and vulnerability-testing modules, including SQL Injection, Reflected XSS, and Brute Force.

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TASK 25 Directory & Endpoint Discovery

Aim

To discover and map directories, files, and endpoints of a web application using Burp Suite Community Edition and directory discovery tools in a controlled testing environment.



Map endpoints

The Target → Site map section was used to examine the application's structure. The discovered resources were organized by host and path.

Recommendations:

- Remove unnecessary or obsolete resources from production applications.
- Enforce authentication and authorization on sensitive endpoints.
- Prevent exposure of configuration files, backups, and secrets.
- Restrict administrative interfaces appropriately.
- Monitor suspicious scanning activity.

CONCLUSION

The Week 7 Web Application Vulnerability Assessment and Penetration Testing (VAPT) assignment provided practical experience in identifying and analyzing common web application security vulnerabilities using **Burp Suite Community Edition, Kali Linux, and Damn Vulnerable Web Application (DVWA)** in a controlled laboratory environment.

During this assignment, Burp Suite was configured to intercept and inspect HTTP requests and responses. The application's endpoints, login functionality, input fields, and parameters were explored using the Proxy and Target Site map features. Manual testing was performed using Burp Suite Repeater to examine application behavior and identify potential security weaknesses.

The testing successfully demonstrated **SQL injection and reflected Cross-Site Scripting (XSS)** in the intentionally vulnerable DVWA application. Authentication-related testing and endpoint discovery were also explored. Screenshots and request-response evidence were collected to support the findings and document the testing process.

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THANK YOU

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